TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

CHAPTER 9: SOLUTIONS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Solutions

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

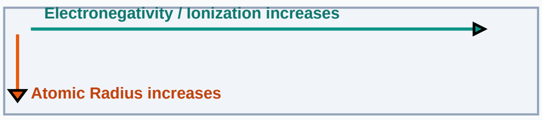
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Henry Law	The solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid.
Raoult Law	The partial vapor pressure of a volatile component in a solution is proportional to its mole fraction.
Colligative Properties	Properties of solutions that depend only on the number of solute particles, not on their nature.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

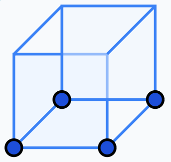
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Simple Cubic Lattice System

Essential Formulas, Laws & Identities:

- Molarity: M = Moles Solute / Volume Solution (L)
- Raoult's Law: p_A = p_A_0 * x_A
- Elevation in Boiling Point: delta_Tb = Kb * m (m = molality)
- Osmotic Pressure: pi = i * C * R * T (i = van 't Hoff factor)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

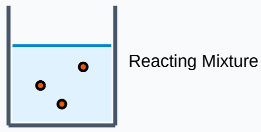
Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find molarity of a solution containing 4g $NaOH$ (Molar mass = 40) in 250ml water.

Step-by-step Solution:
Moles $NaOH$ = 4 / 40 = 0.1 moles.
Volume = 0.25 Liters.
Molarity M = Moles / Vol = 0.1 / 0.25 = 0.4 M.

Titration Beaker & Solute Precipitation



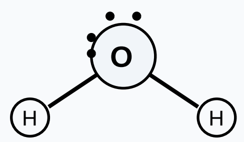
HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Describe non-ideal solutions showing positive and negative deviations from Raoult's law with examples.

Analytical Solution Strategy:
(1) Positive: A-B interactions are weaker than A-A and B-B. Vapor pressure is higher than expected. Example: Ethanol and Acetone.
(2) Negative: A-B interactions are stronger (e.g. hydrogen bonding). Vapor pressure is lower. Example: Chloroform and Acetone.

Lewis Dot Structure (H2O Bent Molecular Geometry)

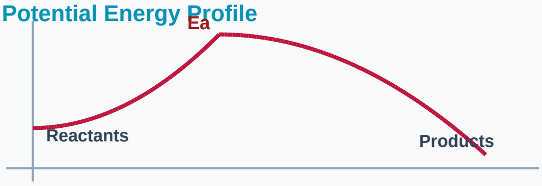


Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

- **Q1 (1-Mark):** State Henry's Law and its limitations.
- **Q2 (2-Mark):** Define molality and mole fraction.
- **Q3 (2-Mark):** Why is elevation of boiling point a colligative property?
- **Q4 (5-Mark):** What is osmosis? Explain reverse osmosis (RO).
- **Q5 (5-Mark):** Calculate osmotic pressure of 0.1M glucose solution at 27C.

Chemical Reaction Potential Energy Profile



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Water Purification

Reverse osmosis desalination plants filter salts using semipermeable membranes.

Application Case: Automotive Systems

Ethylene glycol anti-freeze is added to radiator water to depress freezing points.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C_6H_6) Resonance

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Electrochemical Galvanic Cell System